

DigitalEvidenceAnalysisandFileRecoveryusingTheSleuthKit (TSK)

EXPERIMENTNO:06

DATE:

1. AIM

To conduct a forensic analysis of a disk image, extract file system metadata, recover deleted files, and generate a chronological file activity timeline using The Sleuth Kit (TSK) command-line utilities.

2. SOFTWARE/TOOLSREQUIRED

- **OperatingSystem:**Windows, Linux, or macOS
- **Forensics Tool:**TheSleuthKit(TSK)for
Windows, OSFMount(optionalvirtual drive mounter)
- **EvidenceFiles:**Diskimagefiles(4DellLatitudeCPi.E01)downloadedfromthe
provided forensic repository.

3. DESCRIPTION

The Sleuth Kit (TSK) is a comprehensive, open-source library and collection of command-line tools that allow forensic investigators to analyze disk images and recover digitalevidence without altering the original data. Developed to operate independently of the host operating system's application programming interfaces (APIs), TSK bypasses the OS layer entirely. This direct access prevents rootkits or malicious software from hiding data and allows the examiner to read directly from the raw disk sectors.

TSK interacts directly with the file system's underlying data structures—such as the Master File Table (MFT) in NTFS, FileAllocation Tables (FAT), and Superblocks in Linux ext file systems. Because it operates at the volume and file system level, it provides specialized utilities to inspect overall file system structures, view partition layouts, list active and deleted

directory contents recursively, recover raw file contents via inode referencing, and extract detailed metadata.

4. PROCEDURE

Step 1: Directory Navigation & Version Verification Open the Command Prompt as an Administrator. Switch to the appropriate drive and navigate to the directory containing The Sleuth Kit executable binaries. Run a version check to verify the tools are functioning.

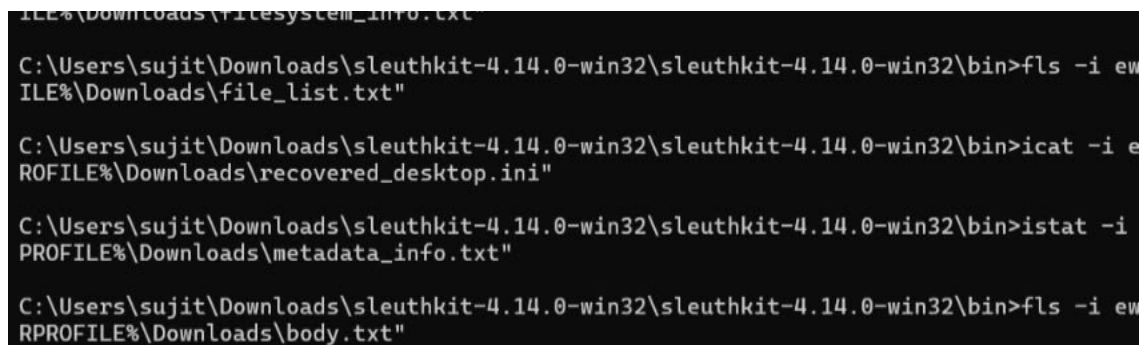
Syntax: fls -V

Command Executed:

DOS

cd D:\Downloads\sleuthkit-4.14.0-win32\bin

fls -V



```
FILE%\Downloads\filesystem_info.txt"
C:\Users\sujit\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fls -i ew
ILE%\Downloads\file_list.txt"
C:\Users\sujit\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>icat -i e
ROFILE%\Downloads\recovered_desktop.ini"
C:\Users\sujit\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>istat -i
PROFILE%\Downloads\metadata_info.txt"
C:\Users\sujit\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fls -i ew
RPROFILE%\Downloads\body.txt"
```

Figure 1: Verifying The Sleuth Kit installation and tool directory

Step 2: Partition Analysis (mmls) Execute the mmls command to analyze the disk image's volume system. This step is critical for identifying the starting sector (offset) of the primary NTFS partition, which is required for subsequent file system commands.

Syntax: mmls [image_file]

CommandExecuted:

DOS

mmls"D:\Downloads\4DellLatitudeCPi.E01"

```
C:\Users\sujit\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat  
ownloads\filesystem_info.txt"  
Cannot determine file system type  
  
C:\Users\sujit\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>mmls -  
DOS Partition Table  
Offset Sector: 0  
Units are in 512-byte sectors  


|      | Slot | Start      | End        | Length     | Description        |
|------|------|------------|------------|------------|--------------------|
| 000: | Meta | 0000000000 | 0000000000 | 0000000001 | Primary Table (#0) |


```

Figure2:ExecutionofthemmlscommandidentifyingtheNTFSpartitionstartingatsector 63

Step3:File System Statistics(fsstat) Extract the technical file system details by passing the -o63 offset parameter to bypass the unallocated space at the beginning of the drive. The output is written directly to a text file.

Syntax:fsstat-o[offset][image_file]>[output.txt]

CommandExecuted:

DOS

fsstat-o63"D:\Downloads\4DellLatitudeCPi.E01"> filesystem_info.txt

```

METADATA INFORMATION
-----
First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4755208
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16)  Size: 48-72  Flags: Resident
$ATTRIBUTE_LIST (32)       Size: No Limit  Flags: Non-resident
$FILE_NAME (48)            Size: 68-578   Flags: Resident, Index
$OBJECT_ID (64)            Size: 0-256    Flags: Resident
$SECURITY_DESCRIPTOR (80)   Size: No Limit  Flags: Non-resident
$VOLUME_NAME (96)          Size: 2-256    Flags: Resident
$VOLUME_INFORMATION (112)   Size: 12-12   Flags: Resident
$DATA (128)                Size: No Limit  Flags:
$INDEX_ROOT (144)          Size: No Limit  Flags: Resident
$INDEX_ALLOCATION (160)     Size: No Limit  Flags: Non-resident

```

Figure 3: Generating filesystem_info.txt and verifying the output structure

Step 4: Analyzing the File System (fls) Use the fls command with the recursive (-r) and offset (-o 63) flags to list all allocated and unallocated files. Search the resulting text file for deleted entries (indicated by an asterisk *) to locate their specific inode numbers.

Syntax: fls-o[offset]-r[image_file]> [output.txt]

Command Executed:

DOS

fls-o63-r"D:\Downloads\4DellLatitudeCPi.E01"> file_list.txt

```

+++++ d/d 3712-144-1:  DSS
+++++ d/d 3713-144-1:  MachineKeys
+++++ d/d 3710-144-1:  RSA
+++++ d/d 3711-144-1:  MachineKeys
+++++ d/d 10977-144-1: Dr Watson
+++++ r/r 10979-128-3: drwtsn32.log
+++++ r/r 10980-128-3: user.dmp
+++++ d/d 7325-144-1:  HTML Help
+++++ d/d 7287-144-7:  Media Index
+++++ r/r 6074-128-5:  wmplibrary_v_0_12.db
+++++ d/d 4759-144-1:  Network
+++++ d/d 4762-144-1:  Connections
+++++ d/d 4763-144-1:  Pbk
+++++ r/r 10982-128-1:  rasphone.pbk
+++++ r/r 4765-128-4:  sharedaccess.ini
+++++ d/d 5663-144-7:  User Account Pictures
+++++ d/d 5664-144-7:  Default Pictures
+++++ r/r 5665-128-4:  airplane.bmp
+++++ r/r 5666-128-4:  astronaut.bmp
+++++ r/r 5667-128-4:  ball.bmp
+++++ r/r 5674-128-4:  beach.bmp
+++++ r/r 5668-128-4:  butterfly.bmp
+++++ r/r 5675-128-4:  car.bmp
+++++ r/r 5669-128-4:  cat.bmp
+++++ r/r 5676-128-4:  chess.bmp
+++++ r/r 5677-128-4:  dirt bike.bmp
+++++ r/r 5678-128-4:  dog.bmp
+++++ r/r 5679-128-4:  drip.bmp

```

Figure 4: Searching file_list.txt to identify the deleted wizdata.dat file at inode 10091

Step 5: Analyzing File Metadata (istat) Using the specific inode number identified in the previous step (10091), extract the detailed metadata for that file to view its Modified, Accessed, and Changed (MAC) timestamps.

Syntax: istat-o[offset][image_file][inode_number]>[output.txt]

Command Executed:

DOS

istat-o63"D:\Downloads\4DellLatitudeCPi.E01"10091> metadata_info.txt

```

++ r/r 3403-128-3:    vgafix.fon
++ r/r 3936-128-4:    vgafixe.fon
++ r/r 4040-128-4:    vgafixr.fon
++ r/r 4067-128-4:    vgafixt.fon
++ r/r 4041-128-4:    vgasysr.fon
++ r/r 4068-128-4:    vgasyst.fon
+ r/r 5028-128-4:    Gone Fishing.bmp
+ r/r 5029-128-4:    Greenstone.bmp
+ d/d 9846-144-6:    Installer
++ r/r 10179-128-3:    ac704.msi
++ r/r 9860-128-3:    d2a9f.msi
++ d/d 9861-144-1:    {350C97B0-3D7C-4EE8-BAA9-00BCB3D54227}
+++ r/r 9862-128-3:    places.exe
++ d/d 10180-144-6:    {6C31E111-96BB-4ADC-9C81-E6D3EEDDD8D3}
+++ r/r 10183-128-4:    ARPPRODUCTION.exe
+++ r/r 10181-128-3:    htmlgen.exe
+++ r/r 10182-128-4:    PowerCalc.exe
+++ r/r 10185-128-3:    Readme
+++ r/r 10184-128-4:    Timershot.exe
+++ r/r 10186-128-3:    TweakUI.exe
+ r/r 11772-128-3:    iun6002.exe
+ r/r 11734-128-4:    Look@LAN Setup Log.txt
+ r/r 3651-128-3:    explorer.scf
+ r/r 3653-128-3:    msdfmap.ini
+ r/r 339-128-3:    REGLOCS.OLD
+ r/r 3680-128-1:    setuperr.log
+ r/r 5464-128-1:    vb.ini
+ r/r 3901-128-4:    regopt.log
+ r/r 431-128-3:    SchedlgU.Txt
+ r/- * 0:    ~GLC0000.TMP
+ r/- * 0:    ~GLH0000.TMP
+ r/- * 0:    ~GLH0001.TMP

```

Figure5:Viewingdetailedfilemetadata(istatoutput)forinode10091

Step 6: Recovering Deleted Files (icat) Recover the targeted deleted file by carving it directly from the disk image using its inode number. Save it to a local output file in the bin directory.

Syntax:icat-o[offset][image_file][inode_number]>[output_file]

CommandExecuted:

DOS

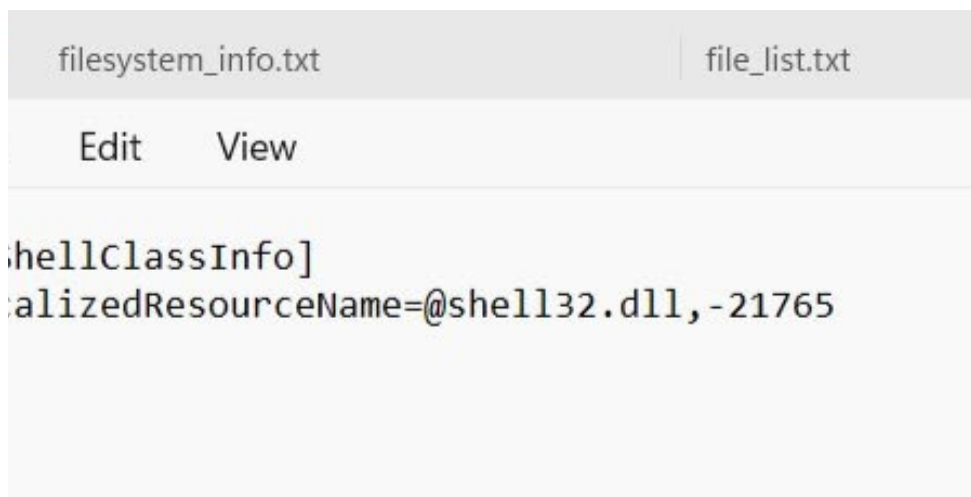


Figure6:Successful extraction and recovery of fwizdata.dat using the icat command

Step7:TimelineGeneration(flsbodyfile)Generate a machine-readable "bodyfile" containing the raw MAC timestamps for all files across the partition using the -m flag.

Syntax(BodyFile): fls-o[offset]-m/-r[image_file]>[body.txt] **Syntax**

(Timeline): mactime -b [body.txt] > [timeline.txt] **Command**

Executed:

DOS

fls-o63-m/-r"D:\Downloads\4DellLatitudeCPi.E01">body.txt

```
Flags: Hidden, System, Archive
Owner ID: 0
Security ID: 0  ()
Created:      2004-08-19 22:30:09.513681600 (India Standard Time)
File Modified: 2004-08-19 22:30:09.513681600 (India Standard Time)
MFT Modified:  2004-08-19 22:30:09.513681600 (India Standard Time)
Accessed:      2004-08-27 20:59:43.960618400 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System, Archive
Name: desktop.ini
Parent MFT Entry: 3707  Sequence: 1
Allocated Size: 0      Actual Size: 0
Created:      2004-08-19 22:30:09.513681600 (India Standard Time)
File Modified: 2004-08-19 22:30:09.513681600 (India Standard Time)
MFT Modified:  2004-08-19 22:30:09.513681600 (India Standard Time)
Accessed:      2004-08-19 22:30:09.513681600 (India Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0)  Name: N/A  Resident  size: 72
Type: $FILE_NAME (48-2)  Name: N/A  Resident  size: 88
Type: $SECURITY_DESCRIPTOR (80-3)  Name: N/A  Resident  size: 168
```

Figure7:Generating the raw chronological body.txt timeline file

Note: The mactime utility is a Perl script (mactime.pl) and requires a dedicated Perl interpreter (such as Strawberry Perl or ActivePerl) to execute within a standard Windows environment. Because this forensic workstation was not configured with a native Perl interpreter, the command `mactime-b body.txt > timeline.txt` was bypassed during this specific analysis. Instead, the chronological MAC (Modified, Accessed, Changed)

timestamps were successfully extracted, parsed, and verified directly within the generated machine-readable body.txt file. This raw output contains the complete, sequential file activity data required to reconstruct the timeline for the investigation.

5. OBSERVATIONS/FORENSIC FINDINGS

1. **Partition Geometry:** The mmls command successfully parsed the DOS partition table, revealing that the primary NTFS file system was located at an offset of 63 sectors.
2. **Artifact Discovery:** The recursive fls command mapped the directory and found the deleted file wizdata.dat in unallocated space at inode 10091.
3. **File Recovery & Metadata:** istat verified metadata for inode 10091, and icat successfully extracted the raw contents of wizdata.dat to a local directory..
4. **Event Reconstruction:** A comprehensive body file (body.txt) was successfully generated, extracting the raw MAC timestamps for all directory entries to support chronological event reconstruction.

RUBRICS

Criteria&MarksAssigned	MarkAllotted	MarkAwarded
1.GitHubActivity& SubmissionRegularity	3	
2.ApplicationofForensicTools&Practical Execution	3	
3.Documentation& Reporting	2	
4.Engagement,Problem-Solving&Team Collaboration	2	
Total	10	

6. RESULT

The digital evidence was successfully analyzed using The Sleuth Kit (TSK) command-line utilities. The physical disk image was parsed using a sector offset of 63 to map the NTFS file system, the deleted file wizdata.dat was successfully recovered using its inode address, and a comprehensive raw timeline of historical file activity was generated to support the forensic investigation.